

Human Review and Escalation Policy

For AI Defect Detection and Root-Cause Assistant

MVTec AD Category Set: bottle, cable, screw, transistor, tile, toothbrush, zipper, metal_nut

Field	Details
Document type	Quality assurance policy / human-in-the-loop escalation guide
Purpose	Define when the AI inspection system must request human review, escalate to a supervisor, or stop automatic decision-making.
Use in RAG	This document is designed for retrieval-augmented generation. It uses clear section titles, keywords, decision tables, and category-specific rules.
Status	Synthetic training and demonstration document for a student project.
Version	1.0
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Important Safety Note

This policy does not allow the AI system to make final safety-critical decisions without human oversight. The AI assistant provides decision support only. Final acceptance, rejection, quarantine, or process shutdown decisions must be confirmed by an authorized human inspector or supervisor when escalation conditions are met.

1. Purpose and Scope

This Human Review and Escalation Policy defines the conditions under which the AI inspection assistant must stop automatic decision-making and involve a human quality inspector, supervisor, process engineer, safety officer, or maintenance team. It is intended for an industrial quality inspection prototype using MVTec AD categories and a RAG-based explanation layer.

The policy supports the following project modules: image quality validation, computer vision anomaly detection, OCR, RAG retrieval, LLM explanation, report generation, and agent routing. It is written to be ingested into a vector database so that the assistant can retrieve the correct escalation rules when answering user questions.

1.1 Covered Product Categories

- Bottle - molded or transparent container inspection.
- Cable - cable harness, insulation, connector, and color alignment inspection.
- Screw - thread, shaft, head, and tip inspection.
- Transistor - electronic component body, legs/pins, alignment, and surface inspection.
- Tile - ceramic or surface tile inspection.
- Toothbrush - handle, bristle, and head inspection.
- Zipper - teeth, slider, fabric, and alignment inspection.
- Metal nut - thread, hole, surface, machining, and contamination inspection.

1.2 Human Review Principle

The AI system may support ACCEPT, REWORK, REJECT, QUARANTINE, RETAKE_IMAGE, HUMAN_REVIEW, or ESCALATE_PROCESS recommendations. However, whenever a rule in this policy is triggered, the system must clearly mark the case as requiring human review and must not present the recommendation as a final autonomous decision.

1.3 Policy Priority

When rules conflict, the safer rule must be followed. For example, if a defect appears minor but model confidence is low, the low-confidence rule triggers human review. If the defect appears critical, the critical severity rule triggers supervisor escalation even when confidence is high.

2. Review Roles and Responsibilities

Role	Main responsibility	When involved
Operator	Captures images, checks basic output, follows non-critical system instructions, records inspection notes.	Normal inspections, retake image requests, simple rework routing.
Quality Inspector	Confirms defect status, severity, and acceptance/rejection decision.	Low confidence, unclear image, major defects, unclear RAG evidence, repeated similar defects.
QA Supervisor	Approves critical rejections, batch quarantine, and escalation to production teams.	Critical defects, safety concerns, repeated failures, potential batch-level impact.
Process Engineer	Investigates possible process causes and production parameter issues.	Repeated category-specific defects, suspected machine setup, tool wear, alignment, heating, cooling, contamination, or assembly cause.
Maintenance Team	Checks machine condition, tooling, sensors, fixtures, and production equipment.	Suspected machine fault, repeated mechanical damage, abnormal contamination, fixture misalignment.
Safety Officer	Assesses worker, consumer, or equipment safety risk.	Sharp edges, electrical safety issue, contamination risk, critical product integrity defect.

Role	Main responsibility	When involved
Data/ML Engineer	Reviews model performance, data drift, false positives, false negatives, and retraining needs.	Model uncertainty, frequent low confidence, unexpected defect type, model mismatch, dataset drift.

3. Global Human Review Triggers

The following triggers apply to every MVTec AD category. If any trigger is true, the system must flag HUMAN_REVIEW or ESCALATE_PROCESS depending on severity.

Trigger ID	Condition	Required system action	Responsible person
HR-001	Model confidence is below 70 percent.	Set HUMAN_REVIEW. Do not finalize accept/reject.	Quality Inspector
HR-002	Model confidence is between 70 and 85 percent and the predicted severity is Major or Critical.	Set HUMAN_REVIEW and show the uncertainty warning.	Quality Inspector
HR-003	Image quality check returns RETAKE_IMAGE.	Request a new image before defect decision. If repeated, escalate to inspector.	Operator / Quality Inspector
HR-004	Image is blurry, too dark, overexposed, strongly reflective, cropped, or product category is unclear.	Set RETAKE_IMAGE or HUMAN_REVIEW depending on condition.	Operator
HR-005	The predicted defect is Critical.	Set ESCALATE_PROCESS. Require supervisor approval.	QA Supervisor
HR-006	The RAG system cannot retrieve relevant evidence for a recommended action.	Set HUMAN_REVIEW. Tell user evidence is missing.	Quality Inspector
HR-007	OCR extracts unreadable or conflicting batch/label/serial data.	Set HUMAN_REVIEW for label verification.	Quality Inspector
HR-008	Detected anomaly is outside the trained or documented defect categories.	Set HUMAN_REVIEW and record unknown defect.	Quality Inspector / ML Engineer
HR-009	Three or more similar defects occur in the same batch, shift, or machine station.	Set ESCALATE_PROCESS and recommend batch sampling.	QA Supervisor / Process Engineer
HR-010	Any safety, electrical, sealing, sharp-edge, contamination, or structural integrity concern is present.	Set ESCALATE_PROCESS immediately.	QA Supervisor / Safety Officer

The LLM must not hide these conditions. If a human review trigger is active, the final answer must explicitly include the phrase: Human review required.

4. Severity, Confidence, and Escalation Matrix

Severity	Confidence $\geq$ 85%	Confidence 70%-84%	Confidence $<$ 70%
Normal	ACCEPT if image quality is PASS and no RAG conflict.	HUMAN_REVIEW if image quality warning exists; otherwise ACCEPT_WITH_NOTE.	HUMAN_REVIEW. Possible false normal or unclear input.
Minor	REWORK or ACCEPT_WITH_NOTE according to SOP.	HUMAN_REVIEW before rework decision.	HUMAN_REVIEW and possibly retake image.
Major	REJECT or QUARANTINE according to SOP; inspector confirmation required.	HUMAN_REVIEW before rejection.	HUMAN_REVIEW. Do not reject automatically.

Severity	Confidence $\geq$ 85%	Confidence 70%-84%	Confidence $<$ 70%
Critical	ESCALATE_PROCESS. Supervisor approval required.	ESCALATE_PROCESS. Supervisor and inspector required.	ESCALATE_PROCESS if visual evidence suggests critical risk; otherwise retake and human review.
Unknown	HUMAN_REVIEW. Do not map to a known defect without evidence.	HUMAN_REVIEW.	HUMAN_REVIEW and ML/data review.

4.1 Escalation Action Codes

Action code	Meaning	Use when
ACCEPT	Product can continue.	Normal prediction, high confidence, good image, no policy conflict.
ACCEPT_WITH_NOTE	Product continues but note is saved.	Low-risk visual mark or weak anomaly without functional risk.
REWORK	Product may be repaired or cleaned.	Minor defect with approved rework path.
REJECT	Product should not continue as saleable/acceptable output.	Major defect confirmed by evidence.
QUARANTINE	Hold product or batch for investigation.	Possible batch issue, repeated defect, contamination, safety issue.
RETAKE_IMAGE	User must upload a better image.	Blur, poor lighting, cropped object, reflection, occlusion.
HUMAN_REVIEW	Inspector must confirm decision.	Low confidence, unclear evidence, unknown anomaly, conflicting signals.
ESCALATE_PROCESS	Supervisor/process team must investigate.	Critical defect, repeated defects, safety risk, process instability.

5. AI Agent Routing Rules

The agent must use this policy to decide whether to answer directly, call RAG, call OCR, ask for a new image, create a report, or escalate the case. The agent should not produce a confident final action when required inputs are missing.

5.1 Required Routing Logic

The following pseudo-workflow is recommended for implementation:

```

IF image_quality == RETAKE_IMAGE:
    action = RETAKE_IMAGE
    explanation = request clearer image
ELSE:
    run vision_model
    IF confidence < 0.70:
        action = HUMAN_REVIEW
    IF severity == CRITICAL:
        action = ESCALATE_PROCESS
    IF product has visible text or label:
        run OCR
        IF OCR unclear or conflicts with expected product data:
            action = HUMAN_REVIEW
    call RAG for SOP, severity guide, action SOP, and this policy
    IF no relevant evidence returned:
        action = HUMAN_REVIEW
    generate final explanation with active triggers and recommended next step

```

5.2 Mandatory Answer Fields

Field	Required content
Human review status	Required / Not required / Escalated / Retake image required.
Trigger ID	Example: HR-001, HR-005, CAT-BOT-03.

Field	Required content
Reason	Clear reason why human review or escalation is required.
Responsible role	Operator, Quality Inspector, QA Supervisor, Process Engineer, Safety Officer, or ML Engineer.
Allowed next action	Only the action allowed by policy: retake, review, reject, quarantine, escalate, etc.
Evidence status	SOP evidence found / missing / conflicting / not needed.

6. Category-Specific Human Review and Escalation Rules

6.1 Bottle Human Review Policy

Key risk areas: Sealing failure, leakage, product contamination, sharp edge, structural weakness, visual quality failure.

Rule ID	Condition	Required action	Responsible role
CAT-BOT-01	Crack, breakage, split, or sharp edge is visible.	ESCALATE_PROCESS	QA Supervisor / Safety Officer
CAT-BOT-02	Foreign contamination, residue, or unknown material appears inside or on the bottle.	QUARANTINE	Quality Inspector / Safety Officer
CAT-BOT-03	Mouth, neck, cap-contact area, or sealing surface appears damaged.	HUMAN_REVIEW	Quality Inspector
CAT-BOT-04	Anomaly is near a functional sealing region but model confidence is below 85 percent.	HUMAN_REVIEW	Quality Inspector
CAT-BOT-05	Multiple bottle defects appear in the same batch or shift.	ESCALATE_PROCESS	QA Supervisor / Process Engineer

RAG retrieval keywords: bottle crack, bottle contamination, sealing defect, bottle neck damage, bottle breakage, bottle human review, bottle escalation

6.2 Cable Human Review Policy

Key risk areas: Electrical failure, insulation damage, wrong wire order, connector failure, short circuit, safety hazard.

Rule ID	Condition	Required action	Responsible role
CAT-CAB-01	Insulation is cut, exposed, burned, punctured, or peeled.	ESCALATE_PROCESS	QA Supervisor / Safety Officer
CAT-CAB-02	Wire order, color alignment, or connector arrangement appears wrong.	HUMAN_REVIEW	Quality Inspector / Electrical Inspector
CAT-CAB-03	Connector pin, end terminal, or cable end appears damaged or missing.	HUMAN_REVIEW	Quality Inspector
CAT-CAB-04	Cable anomaly may affect electrical safety but confidence is low.	ESCALATE_PROCESS	QA Supervisor / Safety Officer
CAT-CAB-05	Repeated cable defects occur within same batch or station.	ESCALATE_PROCESS	Process Engineer / Maintenance Team

RAG retrieval keywords: cable insulation damage, cable connector defect, cable safety risk, exposed wire, wrong cable alignment, cable human review

6.3 Screw Human Review Policy

Key risk areas: Assembly failure, thread mismatch, torque failure, poor fastening, tool wear indicator, mechanical reliability issue.

Rule ID	Condition	Required action	Responsible role
CAT-SCR-01	Thread is missing, flattened, broken, or heavily deformed.	HUMAN_REVIEW	Quality Inspector

Rule ID	Condition	Required action	Responsible role
CAT-SCR-02	Screw tip or head geometry appears damaged, broken, or unsafe.	HUMAN_REVIEW	Quality Inspector
CAT-SCR-03	Foreign material is stuck in threads or on shaft.	REWORK or HUMAN_REVIEW	Operator / Quality Inspector
CAT-SCR-04	Repeated thread defects are detected.	ESCALATE_PROCESS	Process Engineer / Maintenance Team
CAT-SCR-05	Defect may affect assembly fit or torque performance.	HUMAN_REVIEW	Quality Inspector

RAG retrieval keywords: screw thread defect, screw head damage, screw tip damage, screw torque issue, screw process escalation

6.4 Transistor Human Review Policy

Key risk areas: Electrical failure, missing or bent pin, incorrect mounting, component damage, circuit reliability issue.

Rule ID	Condition	Required action	Responsible role
CAT-TRA-01	Pin or leg is bent, broken, missing, misaligned, or touching another pin.	ESCALATE_PROCESS	Quality Inspector / Electrical Inspector
CAT-TRA-02	Body crack, burn mark, chip, or structural damage is visible.	ESCALATE_PROCESS	QA Supervisor / Safety Officer
CAT-TRA-03	Component orientation, placement, or printed marking appears wrong.	HUMAN_REVIEW	Quality Inspector
CAT-TRA-04	Anomaly could affect circuit function but RAG evidence is missing.	HUMAN_REVIEW	Quality Inspector
CAT-TRA-05	Repeated transistor defects are detected in a batch.	ESCALATE_PROCESS	Process Engineer / Maintenance Team

RAG retrieval keywords: transistor bent pin, transistor missing leg, transistor body crack, electrical component review, transistor escalation

6.5 Tile Human Review Policy

Key risk areas: Surface quality failure, crack propagation, sharp edge, installation weakness, cosmetic or structural rejection.

Rule ID	Condition	Required action	Responsible role
CAT-TIL-01	Crack, fracture, broken edge, or large chip is visible.	HUMAN_REVIEW	Quality Inspector
CAT-TIL-02	Sharp broken region or structural weakness is suspected.	ESCALATE_PROCESS	QA Supervisor / Safety Officer
CAT-TIL-03	Surface contamination, stain, discoloration, or glaze issue covers a significant area.	HUMAN_REVIEW	Quality Inspector
CAT-TIL-04	Defect crosses more than one region or appears repeated across tiles.	ESCALATE_PROCESS	Process Engineer
CAT-TIL-05	Lighting/glare makes crack decision uncertain.	RETAKE_IMAGE	Operator

RAG retrieval keywords: tile crack, tile chip, tile glaze defect, tile surface contamination, tile escalation, tile retake image

6.6 Toothbrush Human Review Policy

Key risk areas: Consumer safety risk, hygiene issue, sharp plastic, missing bristles, bristle contamination, usability defect.

Rule ID	Condition	Required action	Responsible role
CAT-TOO-01	Bristles are missing, contaminated, deformed, or uneven in a way that affects use.	HUMAN_REVIEW	Quality Inspector
CAT-TOO-02	Sharp plastic edge, broken handle, cracked head, or loose part is visible.	ESCALATE_PROCESS	QA Supervisor / Safety Officer
CAT-TOO-03	Foreign material or hygiene concern is visible.	QUARANTINE	Quality Inspector / Safety Officer
CAT-TOO-04	Defect may affect consumer use but model confidence is below 85 percent.	HUMAN_REVIEW	Quality Inspector
CAT-TOO-05	Repeated bristle or molding defects are detected.	ESCALATE_PROCESS	Process Engineer

RAG retrieval keywords: toothbrush bristle defect, toothbrush contamination, broken toothbrush head, toothbrush safety escalation, hygiene review

6.7 Zipper Human Review Policy

Key risk areas: Functional failure, missing teeth, slider misalignment, fabric tear, jamming, customer usability defect.

Rule ID	Condition	Required action	Responsible role
CAT-ZIP-01	Missing, broken, bent, or misaligned teeth are visible.	HUMAN_REVIEW	Quality Inspector
CAT-ZIP-02	Slider, puller, or stopper is broken, missing, or badly aligned.	HUMAN_REVIEW	Quality Inspector
CAT-ZIP-03	Fabric tape tear, stitching issue, or foreign material may cause function failure.	HUMAN_REVIEW	Quality Inspector
CAT-ZIP-04	Multiple zipper defects occur in same batch or machine setting.	ESCALATE_PROCESS	Process Engineer / Maintenance Team
CAT-ZIP-05	Image angle prevents tooth alignment evaluation.	RETAKE_IMAGE	Operator

RAG retrieval keywords: zipper missing teeth, zipper slider damage, zipper fabric tear, zipper alignment review, zipper escalation

6.8 Metal Nut Human Review Policy

Key risk areas: Thread fit failure, structural weakness, corrosion, contamination, machining error, assembly reliability issue.

Rule ID	Condition	Required action	Responsible role
CAT-NUT-01	Internal thread damage, blocked hole, or severe deformation is visible.	HUMAN_REVIEW	Quality Inspector
CAT-NUT-02	Crack, fracture, severe dent, or structural defect is visible.	ESCALATE_PROCESS	QA Supervisor
CAT-NUT-03	Rust, corrosion, oil, burr, chip, or foreign material is visible.	REWORK or HUMAN_REVIEW	Operator / Quality Inspector
CAT-NUT-04	Defect may affect fit, torque, or assembly reliability.	HUMAN_REVIEW	Quality Inspector
CAT-NUT-05	Repeated nut machining defects appear in same batch.	ESCALATE_PROCESS	Process Engineer / Maintenance Team

RAG retrieval keywords: metal nut thread defect, metal nut corrosion, nut blocked hole, nut machining defect, nut escalation, metal nut human review

7. Batch-Level and Process-Level Escalation

Human review is not limited to individual images. Some patterns indicate a possible process problem. The system should monitor repeated defect predictions across batch, station, shift, product category, or operator capture session when this metadata is available.

Escalation ID	Pattern	Required action	Reason
BATCH-001	Three or more major defects in same batch.	QUARANTINE batch sample and notify QA Supervisor.	Possible batch-level quality issue.
BATCH-002	Same defect appears repeatedly in the same category within one shift.	ESCALATE_PROCESS to Process Engineer.	Possible machine setting, tooling, material, or handling issue.
BATCH-003	Critical defect appears more than once.	Immediate supervisor escalation and production check.	Possible safety or compliance risk.
BATCH-004	Sudden increase in low-confidence predictions.	Notify ML Engineer and Quality Inspector.	Possible image drift, lighting issue, camera setup issue, or unseen defect type.
BATCH-005	RAG evidence missing for repeated decisions.	Review document knowledge base before continuing automated answers.	The system may be making unsupported recommendations.

7.1 Process Escalation Checklist

- Pause automatic acceptance for affected category or batch until inspector review is completed.
- Collect 5 to 10 representative defect images and 5 normal images for comparison.
- Record product category, batch ID, station ID, shift, operator, timestamp, camera setup, and model version.
- Check whether defect is linked to machine alignment, tooling wear, contamination, material variation, lighting, handling, or packaging step.
- If model drift is suspected, create a data review ticket and do not retrain without labeled evidence.

8. RAG and LLM Safety Rules

The RAG system must retrieve supporting passages before the LLM gives a recommendation that depends on QA policy, severity, corrective action, or escalation. If retrieval fails or returns irrelevant evidence, the assistant must say that the required evidence was not found and must request human review.

Rule	Description	Required wording
RAG-001	No relevant policy passage is retrieved for a defect recommendation.	I could not find enough SOP evidence. Human review required.
RAG-002	Retrieved passages conflict with each other.	The retrieved guidance is inconsistent. Human review required.
RAG-003	Vision output says normal but user reports visible defect.	The AI result conflicts with user observation. Human review required.
RAG-004	LLM suggests possible root cause.	Use possible cause, likely checkpoint, or suspected factor. Do not say confirmed cause unless verified.
RAG-005	Critical defect is detected.	Do not soften the risk. State supervisor escalation clearly.
RAG-006	Evidence is from synthetic QA documents.	Use as project demonstration evidence, not as official factory compliance guidance.

8.1 Forbidden LLM Behaviors

- Do not claim the product is definitely safe when confidence is low or evidence is missing.
- Do not invent a root cause from the image alone.

- Do not ignore human review triggers.
- Do not recommend restarting production after a critical defect without supervisor approval.
- Do not hide uncertainty from the user.
- Do not present synthetic RAG documents as official manufacturer standards.

9. Report Fields Required for Human Review Cases

When a case is flagged for human review or escalation, the generated inspection report must contain enough information for a human inspector to make a decision later.

Report field	Required content
Inspection ID	Unique ID for tracking the case.
Product category	One of bottle, cable, screw, transistor, tile, toothbrush, zipper, or metal_nut.
Image filename	Original uploaded image filename or generated internal file ID.
Image quality result	PASS, PASS_WITH_WARNING, RETAKE_IMAGE, or HUMAN_REVIEW.
Vision result	Normal/defective, anomaly type if available, confidence, severity hint, defect location if available.
Review trigger IDs	All active human review or escalation rule IDs, such as HR-001 or CAT-CAB-01.
Recommended action	RETAKE_IMAGE, HUMAN_REVIEW, REJECT, QUARANTINE, ESCALATE_PROCESS, etc.
Responsible role	Who must act next.
RAG evidence	Document title, section title, and short retrieved evidence snippet.
Uncertainty note	Short note explaining why the AI cannot make a final decision.
Human decision field	Final decision, inspector name, date, and comments should be left for human completion.

9.1 JSON Schema for Implementation

```
{
  "inspection_id": "string",
  "product_category": "bottle | cable | screw | transistor | tile | toothbrush | zipper | metal_nut",
  "image_quality_status": "PASS | PASS_WITH_WARNING | RETAKE_IMAGE | HUMAN_REVIEW",
  "vision_prediction": {
    "defect_status": "normal | defective | uncertain",
    "defect_type": "string",
    "confidence": 0.0,
    "severity": "normal | minor | major | critical | unknown"
  },
  "human_review": {
    "required": true,
    "trigger_ids": ["HR-001"],
    "reason": "string",
    "responsible_role": "string",
    "escalation_level": "operator | inspector | supervisor | process_engineer | safety_officer | ml_engineer"
  },
  "rag_evidence_status": "found | missing | conflicting | not_required",
  "recommended_action": "string",
  "human_final_decision": "pending"
}
```

10. Example Human Review Scenarios

Scenario	AI output	Policy trigger	Final system response
Bottle image shows a crack near neck.	Defective, crack, 91%, critical.	HR-005, CAT-BOT-01, CAT-BOT-03	Escalate to QA Supervisor. Do not accept. Possible sealing/safety risk.
Cable image has exposed wire but confidence is 78%.	Defective, insulation damage, 78%, critical.	HR-002, HR-005, CAT-CAB-01	Escalate process and request electrical/safety review.

Scenario	AI output	Policy trigger	Final system response
Screw thread appears unusual but image is blurry.	Defective, thread damage, 63%, major.	HR-001, HR-003, CAT-SCR-01	Retake image or human review. Do not reject automatically.
Transistor pin appears bent.	Defective, bent pin, 88%, critical.	HR-005, CAT-TRA-01	Escalate to electrical inspector or QA supervisor.
Tile surface has large glare/reflection.	Uncertain, 55%, unknown.	HR-001, HR-004, CAT-TIL-05	Request retake under better lighting.
Toothbrush has foreign material near bristles.	Defective, contamination, 90%, critical.	HR-005, CAT-TOO-03	Quarantine and safety/hygiene review required.
Zipper teeth alignment cannot be judged due to angle.	Uncertain, 68%, unknown.	HR-001, CAT-ZIP-05	Retake image from top-down angle.
Metal nut hole appears blocked.	Defective, blocked hole, 87%, major.	CAT-NUT-01	Human review for fit/assembly risk.

11. RAG Retrieval Keywords

The following keywords should be included in embeddings and test queries so the retriever can find this policy quickly.

human review required, escalation policy, quality inspector review, QA supervisor escalation, low confidence prediction, confidence below 70 percent, critical defect escalation, RAG evidence missing, OCR conflict, image retake required, batch quarantine, process escalation, safety officer review, unknown anomaly, model uncertainty, bottle crack escalation, cable insulation safety, screw thread human review, transistor bent pin escalation, tile crack review, toothbrush contamination quarantine, zipper missing teeth review, metal nut thread defect review

12. Sample RAG Test Questions

- When should the AI assistant request human review?
- What should happen if the defect confidence is below 70 percent?
- Who should review a critical bottle crack?
- What action is required for cable insulation damage?
- Should the LLM give a root cause without evidence?
- What should the system do if no SOP evidence is retrieved?
- What is the escalation rule for repeated defects in the same batch?
- What report fields are required for human review cases?
- What should happen when a tile image has glare and the model is uncertain?
- When should a metal nut defect be escalated to a process engineer?

13. RAG Chunking Guidance

For best retrieval performance, split this policy into chunks of 500 to 900 tokens with 80 to 120 token overlap. Keep rule tables together when possible. Store metadata fields such as document_name, section_title, category, rule_id, severity, action_code, and responsible_role. For category-specific chunks, include the category name in metadata and in the chunk text.

Recommended metadata example:

```
{
  "document_name": "Human_Review_and_Escalation_Policy.pdf",
  "section_title": "Cable Human Review Policy",
  "category": "cable",
  "rule_id": "CAT-CAB-01",
  "action_code": "ESCALATE_PROCESS",
  "responsible_role": "QA Supervisor / Safety Officer"
}
```

14. Implementation Checklist

Checklist item	Required before release
Confidence thresholds implemented	System flags HUMAN_REVIEW below 70 percent and applies uncertainty rules from this policy.
Critical severity logic implemented	Critical defects trigger ESCALATE_PROCESS and supervisor review.
RAG evidence check implemented	System marks HUMAN_REVIEW if evidence is missing or conflicting.
Image retake logic implemented	Poor image quality triggers RETAKE_IMAGE before final defect decision.
Category policy mapping implemented	Each MVTec AD category maps to its specific review rules.
Report schema implemented	Human review fields are saved in generated reports and database records.
LLM safety prompt implemented	The LLM is instructed not to invent root causes or hide uncertainty.
Audit trail implemented	Inspection ID, model version, prompt version, document version, and reviewer decision are stored.

15. Final Policy Statement

The AI Defect Detection and Root-Cause Assistant is a decision-support system. It can identify visual anomalies, retrieve relevant QA knowledge, and recommend next steps. It must not replace human judgment in uncertain, critical, safety-related, undocumented, or repeated process-failure cases. Human review and escalation are mandatory whenever this policy is triggered.